

§4.1 Rotational velocity

$$q_a(t) = R_{ab}(t)q_b \quad \Rightarrow \quad v_{q_a}(t) = \frac{d}{dt}q_a(t) = \dot{R}_{ab}(t)q_b \quad (2.46)$$

$$v_{q_a}(t) = \dot{R}_{ab}(t)R_{ab}^{-1}(t)R_{ab}(t)q_b = \dot{R}_{ab}(t)R_{ab}^{-1}(t)q_a(t) \quad \approx \quad (2.47)$$

Lemma 2.12 (p. 51)

Given $R(t) \in SO(3)$, $\dot{R}(t)R^{-1}(t) \in \mathbb{R}^{3 \times 3}$ and $R^{-1}(t)\dot{R}(t) \in \mathbb{R}^{3 \times 3}$ are both skew-symmetric matrices, $\in so(3)$.

§4.1 Rotational velocity

We consider a rigid body with a body-fixed coordinate frame B , with origin at the orig of the inertial frame, also called the spatial coordinate frame, A . The body, together with the body frame, undergoes rotation only, about an axis through its origin, implying that its origin does not translate.

As a result, point q , away from the origin of frame B and with coordinates q_b in frame B , moves. Of course q_b is constant with time. The coordinates of q in frame A varies with time as $q_a(t) = R_{ab}(t)q_b$.

Instantaneous spatial angular velocity $\omega_{ab}^s \in \mathbb{R}^3$:

$$\hat{\omega}_{ab}^s = \dot{R}_{ab}R_{ab}^{-1} \quad (2.48)$$

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$$\hat{\omega}_{ab}^s = \dot{R}_{ab} R_{ab}^{-1} \quad (2.48)$$

Instantaneous body angular velocity $\omega_{ab}^b \in \mathbb{R}^3$:

$$\hat{\omega}_{ab}^b = R_{ab}^{-1} \dot{R}_{ab} \quad (2.49)$$

$$\hat{\omega}_{ab}^b = R_{ab}^{-1} \hat{\omega}_{ab}^s R_{ab} \quad \text{or} \quad \omega_{ab}^b = R_{ab}^{-1} \omega_{ab}^s \quad (2.50)$$

$$v_{q_a}(t) = \hat{\omega}_{ab}^s(t) q_a(t) \quad (2.51)$$

$$v_{q_b}(t) = \hat{\omega}_{ab}^b(t) q_b \quad (2.52)$$

$$\vec{v}_q = \vec{\omega} \times \vec{q}$$

§4.2 Rigid body velocity

Now we consider the motion of the body and the body fixed frame B , described by the one-parameter curve $g_{ab}(t) \in SE(3)$, relative to frame A , with

$$g_{ab}(t) = \begin{bmatrix} R_{ab}(t) & p_{ab}(t) \\ 0 & 1 \end{bmatrix}$$

$$\dot{g}_{ab}g_{ab}^{-1} = \hat{V}_{ab}^s$$

a twist, called the *spatial velocity*, with twist coordinates

$$V_{ab}^s = \begin{bmatrix} v_{ab}^s \\ \omega_{ab}^s \end{bmatrix} = \begin{bmatrix} -\dot{R}_{ab}R_{ab}^T p_{ab} + \dot{p}_{ab} \\ \left(\dot{R}_{ab}R_{ab}^T\right)^\vee \end{bmatrix} \quad (2.53)$$

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$$v_{q_a} = \hat{\omega}_{ab}^s q_a + v_{ab}^s \quad (\text{2nd of 2.54})$$

$$(3 \times 1)$$

In homogeneous form (4×1) :

$$v_{q_a} = \hat{V}_{ab}^s q_a \quad (\text{1st of 2.54})$$

§4.2 Rigid body velocity

$$g_{ab}^{-1} \dot{g}_{ab} = \hat{V}_{ab}^b$$

a twist, called the *body velocity*, with twist coordinates

$$V_{ab}^b = \begin{bmatrix} v_{ab}^b \\ \omega_{ab}^b \end{bmatrix} = \begin{bmatrix} R_{ab}^T \dot{p}_{ab} \\ \left(R_{ab}^T \dot{R}_{ab} \right)^\vee \end{bmatrix} \quad (2.55)$$

§4.2 Rigid body velocity

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$$v_{q_b} = \hat{\omega}_{ab}^b q_b + v_{ab}^b \quad (\text{2nd of 2.56})$$

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$$v_{q_b} = \hat{V}_{ab}^b q_b \quad (\text{1st of 2.56})$$

§4.2 Rigid body velocity

Adjoint transformation:

$$\text{Ad}_g = \begin{bmatrix} R & \hat{p}R \\ 0 & R \end{bmatrix} \quad (2.58)$$

Top of p. 56: $\text{Ad}_g^{-1} = \text{Ad}_{g^{-1}}$

Lemma 2.13 (p. 56): If $\hat{\xi} \in se(3)$ is a twist, $\xi \in \mathbb{R}^6$, then, for any $g \in SE(3)$, $g\hat{\xi}g^{-1}$ is a twist with twist coordinates $\text{Ad}_g\xi \in \mathbb{R}^6$.

Relation between spatial and body velocities:

$$V^s = \text{Ad}_g V^b \quad (2.61)$$

§4.3 Velocity of screw motion

For case where $g_{ab}(\theta) = e^{\hat{\xi}\theta} g_{ab}(0)$ (eq. (2.45) p. 49):

$$\hat{V}_{ab}^s = \hat{\xi} \dot{\theta}$$

$$\hat{V}_{ab}^b = \left(\text{Ad}_{g_{ab}^{-1}(0)} \hat{\xi} \right)^\wedge \dot{\theta}$$

§4.4 Coordinate transformation

Proposition 2.14 (p. 58) Transformation of spatial velocities:

Consider the motion of three coordinate frames, A , B , and C . The following relation exists between their spatial velocities:

$$V_{ac}^s = V_{ab}^s + \text{Ad}_{g_{ab}} V_{bc}^s$$

Proposition 2.15 (p. 59) Transformation of body velocities:

Consider the motion of three coordinate frames, A , B , and C . The following relation exists between their spatial velocities:

$$V_{ac}^b = \text{Ad}_{g_{bc}^{-1}} V_{ab}^b + V_{bc}^b$$

§4.4 Coordinate transformation

Lemma 2.16 (p. 61):

$$V_{ab}^b = -V_{ba}^s$$

$$V_{ab}^b = -\text{Ad}_{g_{ba}} V_{ba}^b$$